Apply the next mode before lifting reciprocal candidates

T-1864 · 2026-11-07 · Prime-field quadratic quotient algebra

The next sign-mode constraint now acts directly on compressed reciprocal candidates. Balanced streamed evaluation represents $F_1(V)$, $F_2(V)$ as pairs $a(W)+b(W)V$ modulo $G_u(W)$ and V^2-WV+1. No full mode vector or candidate-root enumeration is needed.

Compressed first-mode branch $G_u(W)$, $W = V + V^{-1}$

Stream F_1 and F_2 as pairs $a(W) + b(W)V$

Reduce modulo $G_u(W)$ and $V^2 - WV + 1$; retain phase $P = V^n - u$

$K = \gcd(G_u, \text{norm}(P), \text{norm}(C_2), \text{determinant}(P,C_2))$

Only then: capped lift ⟶ phase filter ⟶ actual C_2 filter

Union + exact zero test + scale ⟶ original recovery state

Same cap7: 40 completions / 235 fallbacks (275 attempts; no timing claim)

Exact orientation compatibility. For $P=a+bV$ and $C=c+dV$, a common quadratic root exists exactly when both norms $a^2+Wab+b^2$ and $c^2+Wcd+d^2$ and the determinant $ad-bc$ vanish. Separate norms can select opposite roots. The proof includes zero leading coefficients and repeated quadratic fibers. After filtering W, a capped lift still enforces the phase and actual C_2; zero is checked separately.

Native exact control	Result
Pair cases / norms-only false positives	7,424 / 12 rejected
Zero-leading / ramified controls	3,248 / 1,536
Mode configurations / exact mode values	18 / 324
Exact full dense intersections	972
Public targets / source calls	281 / 2,881

Same-cap construction result. All public scalars, routes and statistics match the isolated path. At cap7, the new path completes40of275attempts and falls back235times; the preserved predecessor completes10and falls back265. Among890processed branches, projection-degree sum falls from2370to1530, with410strict reductions. Early returns exclude unprocessed branches from these sums. No timing ratio follows.

Charged work. Three base polynomial products per pair multiplication; balanced streamed mode evaluation, phase powers, norms, determinant, squarefree normalization and gcds all count. Modes are reevaluated per branch. Pair-state space is conservatively $O(d \log(L+1)+B)$, plus context/kernel allocations. Charge Vec capacity and failed work before fallback. The lift cap is not a process byte ceiling.

Evidence. validation01 pins57transitive inputs and exact output; native overflow checks pass. Initial generator quoting error was fixed before native compilation; failed source/error retained. Producer evidence unsigned. No cloud/timing run or fullCargo rerun; original4,563-case cohort unchanged. See DERIVATION.md for complete proof and replay.

Sources and next step. Exact predecessor; Bhargava–Zieve reciprocal fibers (websites.umich.edu/ zieve/papers/manjul.pdf); official FLINT modular conventions. KB unavailable; no novelty claim. Next keep phase-tagged states across further radii/history constraints with exact common-orientation handling.